

NARWC Database Project — Status

Branch: main | Last updated: 2026-07-20

1. Current state

The system can extract per-survey CSV files from a monolithic legacy flat-file (SAS-era export), validate each survey against 9 rule modules covering required fields, coordinates, datetime, species, environmental conditions, Beaufort sea state, behavior codes, photos, and foreign-key constraints, acknowledge specific validation warnings via a version-controlled override file (`config/overrides/<batch>_overrides.csv`, e.g. `migration_overrides.csv`), and upload validated surveys to a SQL Server Master table via transaction-safe inserts (rollback on failure). The override system supports per-row acknowledgements keyed on (`fileid`, `eventno`, `field`, `rule_id`) and per-survey acknowledgements (empty `eventno` to suppress a warning class across an entire survey). A second, routine-ingestion pipeline (`scripts/ingestion/`) now shares the same validate+upload code for per-contributor batches going forward. Test suite (as of 2026-07-20, `tests/unit`): 172 passing, 0 failing, 10 incomplete (DB-dependent tests self-skip without a live connection).

What the system cannot do yet: provide a curator-facing GUI, or apply Bob's known bulk corrections automatically. The NEAQFormat parser is implemented as the reference contributor-parser (see `src/+narwc/+io/+parsers/NEAQFormat.m`), but only for the 3 field renames confirmed against `format_definitions.json` — no real NEAQ export file has been available to verify anything beyond that. Live SQL Server transaction behavior has not been verified end-to-end; the Mac-side mock tests verify control flow only.

2. Project scope

The project delivers two things:

1. Legacy migration — migrate the NARWC sightings database from a SAS-era flat-file CSV into a modern SQL schema with validation, error tracking, and a version-controlled correction record.
2. Curation tools — provide a sustainable pathway for ingesting new surveys and a curator-facing interface for managing data quality going forward.

Deadline: end of August 2026. Curator is away August 1–18 and will be the primary user of the curation tools post-handoff. Phase 1 (legacy migration) should be complete before she leaves; Phase 2 tools should be in usable shape on her return.

3. Repository map

```

NARWC-DB/
├─ startup.m          # Adds all paths, checks toolboxes, creates data dirs –
run first
├─ config/
│   ├── load_config.m      # Layered config loader: defaults < local < batch
│   ├── get_config.m       # Legacy config source (paths, validation params); still use
│   ├── get_lookup_table.m # Loads CSV lookup tables from data/tables/ by name
│   ├── reload_config.m    # Clears cached get_config singleton
│   │   └─ defaults/      # Version-
│   │       controlled baseline values (db, validation, pipeline)
│   │   ├── local/        # Gitignored; contains db_config_local.m with credentials
│   │   └─ batches/       # Per-workflow override files (migration.m, future: routine.m)
│   │       └─ overrides/  # Per-batch warning–
│   └─ override CSVs and README
├─ src/
│   ├── +migration/
│   │   └─ apply_known_fixes.m # Category C corrections applied pre-
validation
│   │   ├── ConversionValidator.m # FIXME:DELETE – unused
│   │   └─ MetadataExtractor.m   # FIXME:DELETE – unused
│   └─ +narwc/
│       ├── +ingestion/
│       │   └─ BatchUploader.m    # Workhorse: validates + uploads from pending/, tr
safe overwrite
│   └─ SurveyExtractor.m          # Splits monolithic legacy CSV into per-
survey CSV files
│       └─ SurveyFileWriter.m     # Shared per-
FILEID chunk writer, used by SurveyExtractor and convert_contributor_batch
│           ├── convert_contributor_batch.m # Routine ingestion: parses a contributor's ra
│           └─ run_batch_upload.m          # Shared connect-
>upload->stats->close, used by both pipelines
│               └─ +db/
│                   ├── Connection.m      # DB connection wrapper; includes beginTransaction/commi
│                   └─ FieldDefinitions.m # Single source of truth for all 55 field names and type
│               └─ +io/
│                   ├── DataTypeConverter.m # Type coercion before sqlwrite()
│                   └─ +parsers/

```



```

└─ sanitize_filename.m

└─ scripts/
  └─ smoke_validate.m          # Quick end-to-
end validation smoke test (TODO: move to tests)
  └─ migration/                # One-time legacy migration
    └─ validate_csv_database_lines.m  # Step 0: per-
line CSV sanity check
    └─ step1_extract_surveys.m      # Step 1: calls SurveyExtractor
    └─ step2_upload_surveys.m       # Step 2: calls shared run_batch_upload
    └─ step3_validate_migration.m    # Step 3: analyzes results, generates report
    └─ run_full_migration.m         # Runs steps 1-
3 in sequence
    └─ generate_migration_report.m   # Produces markdown/HTML report + charts
    └─ verify_migration_results.m    # Reconciles DB vs. split-
summary source; FK integrity checks
  └─ ingestion/                # Routine, per-
contributor-season ingestion
  └─ convert_contributor_batch.m     # Resolves contributor parser, splits by FILEID
  └─ upload_contributor_batch.m      # Validates + uploads via shared run_batch_upload
  └─ sql/                        # T-
SQL schema and operational scripts – see scripts/sql/README.md
  └─ schema/                    # 01-
06: create DB → tables → indexes → FKs → populate lookups
  └─ verification/              # Row counts, FK integrity checks
  └─ curation/                  # delete_survey, find_duplicates, recent_uploads
  └─ migration/                 # apply_known_fixes (SQL fallback for post-
upload correction)
  └─ teardown/                  # drop_all_tables, truncate_master (dev only)
  └─ setup/
    └─ test_connection.m         # Quick DB connection test
    └─ pull_lookup_tables.m      # Pulls lookup tables from DB into local CSVs
    └─ push_lookup_tables.m      # Pushes local CSV lookup tables into DB

└─ tests/
  └─ test_runner.m / run_*.m     # Test runner infrastructure
  └─ fixtures/
    └─ TestFixtures.m           # Generates and loads mock survey data
    └─ README.md                # (planned)
    └─ sample_data/             # Per-
survey fixture CSVs (aT*, fT*, oT*, pT*, HT*)
    └─ aT99001_volume.csv       # Synthetic high-
volume fixture for smoke tests

```

```

└─ unit/
    └─ test_validation.m
    └─ test_parsers.m
    └─ test_processing.m
    └─ test_reports.m
    └─ test_db_connection.m          # Requires live DB
    └─ test_get_config.m
    └─ test_characterization_batch.m
    └─ test_characterization_extractor.m
    └─ test_characterization_parser.m
    └─ test_upload_guardrail.m

└─ lib/
    └─ +logging/                    # Logging toolbox (Logger class + level functions)

└─ data/                          # Runtime data (mostly gitignored)
    └─ overrides.example.csv        # Example/template for overrides
    └─ tables/                     # Lookup table CSVs (committed)
    └─ legacy/                     # Source CSVs from SAS export
    └─ raw/pending|processed|rejected/ # Staging dirs for new survey ingestion

└─ docs/                          # Architecture docs, rule guide, testing guide, override
└─ scripts/sas/                   # SAS QC scripts (Chk*.sas) + legacy PRG/DBF files
└─ ref/                           # NARWC Users Guide PDF and other reference material

```

4. Active work

- SQL Server schema deployment — all 6 schema scripts written (scripts/sql/schema/); next step is filling in the <FILL_IN> path in 06_populate_lookup_tables.sql and running scripts 01–06 against the SQL Server instance. See handoffs/db_setup_handoff.md.
- Personal pipeline walkthrough (Russ) — run the full migration end-to-end on a local copy to surface remaining blockers; not yet started
- Lookup table updates (Category A, 19 codes) — add missing platform codes (13), species codes (5), behavior codes (2), ANHEAD codes (2), BLOCK code (MB), GLARE code (9) to data/tables/; prerequisite for most remaining FK validation failures; pending Bob confirmation on each entry. See §8.4.
- ANHEAD lookup expansion — clarify whether ANHEAD=19 is valid or a sentinel; requires domain expert input
- Per-survey corrections — specific event/field fixes for individual surveys; see §8.5. Prerequisite for those surveys passing validation.

- **apply_known_fixes.m** (Category C) — done. `src/+migration/apply_known_fixes.m` implements all 8 fixes; called by `BatchUploader.uploadFromFolder` between CSV parse and validation, gated by `config.pipeline.known_fixes.enabled`. SQL script retained as post-upload fallback. See `docs/known_fixes.md`.
 - Coordinate warning messages — done (2026-06-29). `coordinate_rules.m` now includes the actual lat/lon value in all out-of-range and outside-survey-area warning messages.
 - Package layout refactor — rename `+io/` → `+ingestion/` (parsers), move `BatchUploader` to `+db/`, extract single-survey Uploader from `BatchUploader`; pre-August handoff goal
 - SAS rule porting — port SAS QC checks (`scripts/sas/Chk*.sas`) to MATLAB validation rules; TAXCODE-aware NUMBER thresholds implemented; remaining SAS checks TBD
 - Opportunistic sighting field warnings — `required_fields.m` raises 'error' for missing DAY/MONTH/TIME; these should be 'warning' for opportunistic sightings (identified by survey type prefix or explicit flag). Not yet implemented.
-

5. Recent commits

2026-06-29 Add `typical_max_group/typical_max_calf` to SPECCODE and TAXCODE; validator NUMBER/NUMCALF rules now use SPECCODE → TAXCODE → global threshold cascade. Curators adjust thresholds via CSV, no code change. Commits A+B: schema/CSV/SQL migration + `species_rules.m` rewrite.

2026-06-26 ade7fe1 debugging some sql stuff

2026-06-26 875f69c Updated the tests and files to match actual surveys and pass tests

2026-06-26 bb65c5e Add `scripts/sql/` scaffolding for DB-side scripts

2026-06-26 10b275a Add pipeline walkthrough doc for curator onboarding

2026-06-26 abb08f6 Refactor `PROJECT_STATUS.md` for readability and maintainability

2026-06-26 SQL schema scripts fully written: all 6 schema files, `docs/database_schema_scripts/sql/README.md`, `handoffs/db_setup_handoff.md`. Fixed USE NARWC → USE across all non-schema scripts. Fixed BLOCK check bug in `check_fk_integrity`

6. Deferred / future work

- Phase B pattern overrides (`fileid_pattern` glob matching) — allows a single override entry to suppress a warning across multiple surveys matching a pattern; extension point stubbed in `SurveyValidator.buildMatcherList()`; defer until real curator use surfaces the need

- GUI integration with override workflow — Curator-friendly interface over the current CSV-based override mechanism; post-August
- Curator-friendly bypass UX — simpler interface for the curator over the existing override/ingestion machinery; post-August
- NEAQFormat parser and new-survey ingestion pathway — Phase 2b; NEAQFormat.m is currently a stub
- Manual correction tool (apply_manual_correction.m) — interactive tool for Bob's one-off data corrections; post-August
- Historical SAS bulk corrections documentation (docs/historical_corrections.md) — inventory what the PRG/DBF files in scripts/sas/ historically did; out of scope to port
- Live SQL Server transaction verification — must verify beginTransaction / commit / rollback behavior with the JDBC driver before the next production migration run
- Codetag cleanup pass — review and fix FIXME/TODO comments throughout src/ and scripts/; post-August

For the current TODO/FIXME/NOTE inventory, run:

```
grep -rn 'TODO\|FIXME\|NOTE' src/ scripts/ tests/ --
include='*.m'
```

7. Open questions

- **required_fields.m** accuracy: default_config() lists DDSOURCE, EVENTNO, FILEID, IDSOURCE, YEAR; central config (get_config) lists LAT_DD, LONG_DD, YEAR, MONTH, DAY. Neither is confirmed against the database schema NOT NULL constraints. Blocks confidence in the required-field validation rule.
- **SurveyValidator** config split: Resolved. load_config('migration') returns permissive thresholds for the legacy migration; load_config() returns strict defaults for routine ingestion. Override CSV path comes from the batch config. validation_config.m deleted.
- **visibility_allow_negative** split: BatchUploader sets this to false (strict) by default, true only in LegacyMode. The global get_config default remains true for backwards compatibility. Callers that construct SurveyValidator directly still get the permissive value.
- **temporal_rules.m** filename typo: Will cause issues on case-sensitive filesystems; rename pending.
- ANHEAD=19 intent: README says “ANHEAD = 19 is a valid error” — ambiguous. Clarify before adding to lookup table.
- Live SQL Server transaction semantics: beginTransaction / commit / rollback verified only with Mac-side mock. Production behavior with the JDBC driver in use is unconfirmed. Blocks confidence in the transaction-safe overwrite for production migration runs.

8. Reference

8.1 Architecture and design decisions

Validator thresholds are data-driven. NUMBER and NUMCALF thresholds for group-size warnings cascade: `SPECCODE.typical_max_group` → `TAXCODE.typical_max_group` → config global default (1000 / 100). Curators adjust thresholds by editing `data/tables/SPECCODE.csv` or `TAXCODE.csv` and running `push_lookup_tables.m` — no MATLAB code change needed.

Warning override philosophy. Validation warnings block upload unless explicitly acknowledged. Acknowledgements are stored in `config/overrides/<batch>_overrides.csv` (version-controlled), keyed by (fileid, eventno, field, rule_id). A per-row entry suppresses exactly one warning instance. A per-survey entry (empty eventno) suppresses a (fileid, field, rule_id) combination across the entire survey. The `AllowWarnings = true` flag is preserved as an emergency escape valve but is not the intended workflow. Curator workflow documented in `docs/warning_overrides.md`.

Data correction philosophy. Preserve what was originally recorded; do not silently transform data. Manual corrections are opt-in, audited, and tracked via `overrides.csv` or an explicit correction script. Speed and climb-rate threshold violations are warnings only (data recorded; threshold is advisory). The migration does not clean data in flight — it validates and rejects, forcing explicit correction decisions.

Package layout (current state). `+ingestion/` holds the write-side workhorses — `BatchUploader`, `SurveyExtractor`, `SurveyFileWriter`, `convert_contributor_batch`, `run_batch_upload` — shared by both the migration and routine-ingestion pipelines (see CLAUDE.md's "Ingestion Pipelines" section). Parsers stayed in `+io/+parsers/` rather than moving under `+ingestion/`; no separate single-survey Uploader was extracted from `BatchUploader`, and `BatchUploader` did not move to `+db/`. `+validation/` and `+processing/` are unchanged. See §8.3 for the relocation history that did land.

8.2 Known bugs

BLOCK column type varies across surveys (latent — not surfaced by current tests).

When two survey tables loaded with plain `readtable` are combined with `vertcat`, MATLAB may throw "Cannot concatenate the table variable 'BLOCK' because it is a cell in one table and a non-cell in another." Cause: `readtable` without explicit import options infers column type per file. Files with only empty

BLOCK values yield double (all NaN); files with alphanumeric codes yield cell (R2020b) or string (R2021a+). StandardFormat uses delimitedTextImportOptions to enforce types but any caller that bypasses the parser inherits this behavior.

Mitigation (2026-06-24): buildCombined helper in test_characterization_extractor.m detects any variable whose class diverges across input tables and coerces to cell-of-char before vertcat. The underlying StandardFormat type-enforcement gap is still open; address when adding a multi-file parser characterization test.

8.3 Historical: refactor branch progress

The major structural moves completed on the refactor branch:

BatchConverter → BatchUploader (2026-06-24). src/+migration/BatchConverter.m relocated to src/+narwc/+ingestion/BatchUploader.m. Class renamed to narwc.ingestion.BatchUploader. Constructor gains TableName (default 'Master') and LegacyMode (default false) options. LegacyMode = true sets visibility_allow_negative = true in the validator config. step2_upload_surveys.m updated to pass 'LegacyMode', true.

SurveyExtractor relocation (2026-06-24). src/+migration/SurveyExtractor.m relocated to src/+narwc/+ingestion/SurveyExtractor.m. All call sites in scripts and tests updated; git grep migration.SurveyExtractor returns zero hits.

Transaction-safe overwrite (2026-06-24). BatchUploader.uploadSurvey() now wraps delete + insert in a database transaction. beginTransaction() / commit() / rollback() methods added to narwc.db.Connection. Prior AutoCommit state saved and restored. If the driver does not support transactions, a warning is logged and the operation proceeds non-atomically. Four characterization tests in test_characterization_batch.m verify control flow. Live DB verification still required.

Warning override system (2026-06-25). data/overrides.csv introduced. ErrorCollector stores rule_id and eventno on every entry. All 9 rule modules emit stable rule_id values. SurveyValidator loads overrides at construction and demotes matched warnings to info. _errors.log and _run_summary.csv both append across runs.

Per-survey override mode (2026-06-25). Empty eventno in overrides.csv acknowledges all warnings of a (fileid, field, rule_id) combination across an entire survey. Results expose warnings_acknowledged_per_row and warnings_acknowledged_per_survey.

8.4 Open validation/code tasks

Items surfaced during migration that require code or config changes (not yet implemented):

Task	Notes
Opportunistic fields (DAY/MONTH/TIME) → warning	required_fields.m raises error; should be warning for opportunistic sightings
ANHEAD > 22 → NaN	Needs a new fix in apply_known_fixes.m (or post-upload SQL); threshold from field manual
Check year/month/day matches FILEID	New validation rule; FILEID encodes the survey date — mismatch indicates data-entry error
f403158: GLARER=7 → 1	One-off correction; add to apply_known_fixes.m or override CSV
o105921: PLATFORM=164 → 900	Unknown platform code; 900 is the “other/unknown” sentinel; confirm with Bob
o112971: DDSOURCE unknown	May be resolved by copying IDSOURCE to DDSOURCE; confirm with Bob
o121911: CONFIDNC row 444 = 90 → 0	Invalid code; 0 is the correct value
o123921: EVENT 159 TIME=003716/DAY=11; EVENT 211 TIME=001240/DAY=26; EVENT 212 TIME=005201/DAY=26; EVENT 283 LONG_DD=-71.93525 (missing negative sign)	Four discrete corrections; add to override CSV or apply_known_fixes

8.5 Lookup table audit

Status as of 2026-06-23; refresh after lookup table updates land.

All entries below are absent from their respective CSVs. Adding them and running scripts/setup/push_lookup_tables.m is the prerequisite for resolving the majority of FK validation failures in the legacy data. Pending Bob confirmation on each entry.

Table	Missing entries
SPECCODE.csv	CV-S, CV-O, CV-P, CV-R (wind-farm construction vessels), ECOT (dolphin watching)
Behave.csv	73 (security zone patrol), 74 (pile-driving)
ANHEAD.csv	19 (intent unclear — clarify before adding), 20 (underway, course unknown)
Block.csv	MB (Mass Bay — not in manual; confirm before adding)
GLARE.csv	0 (legacy missing-value sentinel — add as valid code or null-ify existing 9 values in data)
PLATFORM.csv	193, 194, 266, 268, 280, 325, 329, 330, 332, 573, 637, 644 (13 codes absent from PLATFORM.csv)

Platform notes: 573 = towboat/similar; 637 = APEM Partenavia; 266 = Canadian Coast Guard; 193 = Mingan Island Cetacean Study; 280 = misc./unknown Canadian vessel; 325 = Fugro Explorer; 194 = Helen H; 268 = R/V Leeway Odyssey; 329/330/332/644/70 = MMO vessels around wind farm.

8.6 Survey-specific data observations

Known data quality issues in individual surveys. “Correction needed” items require explicit fix via `apply_known_fixes.m`, override CSV, or manual DB correction with Bob’s approval.

Survey Issue	Status / Action
c01810 BEHAV1 values may be placed in ANHEAD field	Investigate
f011048 BEHAV1 values may be placed in ANHEAD field	Investigate
f203360 BEHAV1 values may be placed in ANHEAD field	Investigate
f403158 GLARER=7 should be 1	Needs correction (ap- ply_known_fixes or over-ride)
f607053 BEHAV1 values may be placed in ANHEAD field	Investigate
f608034 BEHAV1 values may be placed in ANHEAD field	Investigate
o105921 PLATFORM=164 (unknown code); likely should be 900 (other/unknown)	Confirm with Bob

Survey Issue	Status / Action
o11297 IDSOURCE unknown; may be copied from IDSOURCE	Confirm with Bob
o11392 PLATFORM =266 = Canadian Coast Guard (add to lookup — see §8.5)	Lookup table gap
o11700 Lat outside typical survey area and early year are both valid — opportunistic sighting	Add override CSV entry
o11892 Lat /lon outside typical area is valid	Add override CSV entry
o12191 CONFIDNC row 444 = 90 (invalid); should be 0	Needs correction
o12392 EVENT 159: TIME=003716/DAY=11; EVENT 211: TIME=001240/DAY=26; EVENT 212: TIME=005201/DAY=26; EVENT 283: LONG_DD=-71.93525 (missing negative sign)	Four corrections needed
p31272 EVENT 540: LAT_DD=0, LONG_DD=0 (both missing in source data; printout shows interpolated 48.042/-63.714)	Investigate — likely missing, not zero
p90516 Large survey with many comments; suspected data corruption	Needs manual review before migration

9. Glossary

Term	Definition
NARWC	North Atlantic Right Whale Consortium — the organization managing this sightings database
NARW / RIWH	North Atlantic Right Whale; RIWH is the species code used in the database
FILEID	Survey identifier — a short alphanumeric string (e.g., F098027) that groups all event rows from a single survey effort. First character typically indicates survey type (see below).
EVENTNO	Sequential event number within a survey. Not globally unique; unique only within a FILEID.
Master	The primary SQL table into which all survey records are uploaded.
SPECCODE	4-character species code (e.g., RIWH, FIWH, HUWH). Validated against data/tables/SPECCODE.csv.
TAXCODE	Taxonomic category code (numeric); validated against data/tables/TAXCODE.csv.
PLATFORM	Numeric code identifying the vessel or aircraft used for the survey. Validated against data/tables/PLATFORM.csv.
BEHAV1–BEHAV15	Up to 15 behavior codes per event row. Validated against data/tables/Behave.csv.
ANHEAD	Angle to the animal's head (1–8 compass points, or coded values). Not the same as HEADING.
HEADING	Direction of travel of the platform (degrees).
BEAUFORT	Beaufort sea state scale (0–9 standard; 0–12 in full scale).
DDSOURCE	Position data source code (e.g., GPS, radar). FK to DDSOURCE lookup.
IDSOURCE	Identification source (how the sighting was identified). FK to IDSOURCE lookup.
IDREL	Identification reliability code. FK to IDREL lookup.
CONFIDNC	Confidence code for the sighting identification. FK to Confidnc lookup.
LEGNO / LEGTYPE / LEGSTAGE	Survey leg number, type (e.g., on-effort, off-effort), and stage. FK to respective lookup tables.
STRIP	Strip transect identifier/flag. Values > 16 are considered invalid in the legacy data.
BLOCK	Geographic survey block identifier (string). FK to BLOCK lookup.
STRATUM	Survey stratum identifier. FK to STRATUM lookup.
GLAREL / GLARER	Glare intensity on left and right sides of the trackline (0–3 valid; 9 is a legacy missing-value code).
VISIBLTY	Visibility in nautical miles. Negative values occur in legacy data (coded via OLDVIZ lookup table).
SURFTEMP	Sea surface temperature in °C.
WX	Weather code. FK to WX lookup.

Term	Definition
CLOUD	Cloud cover in oktas (0–8). FK to CLOUD lookup.
PHOTOS	Whether photos were taken (1=No, 2=Yes slides/prints, 3=cine, 4=video, 5=multiple). FK to PHOTOS lookup.
NUMCALF	Number of calves observed in the sighting group.
S_LAT / S_LONG / S_TIME	Starting latitude, longitude, and time for the event leg/trackline.
ANGLEL / ANGLER	Angle to sighting left and right of the trackline.
Survey type prefixes	First character of FILEID encodes survey origin: f = aerial fixed-wing, c = aerial (CCS?), o = opportunistic, p = shipborne — based on README examples; not formally documented.
Opportunistic sightings	Sightings not from systematic survey effort; Day/Month/Time fields may legitimately be missing. Should produce warnings, not errors.
Bob	Domain expert referenced in the README who is making manual data corrections to certain surveys.
Curator	Curator who will own the database and curation tools post-handoff (August 2026).
RUSS_24.CSV	The legacy flat-file CSV being migrated (hardcoded in <code>validate_csv_database_lines.m</code> and referenced in <code>run_full_migration.m</code>).